ECOM90009 Quantitative Methods for Business

Workshop 8 (Week 9)

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Introduction

Zheng Fan

- Know more about myself: zhengfan.site
- Ph.D student in Economics at Unimelb

If you need any help, don't be shy

- Weekly consultation & Ed discussion board (your first priority)
- Email: `jnlye@unimelb.edu.au` for subject related issues
- Consult Stop 1 for special consideration
- Email: `fan.z@unimelb.edu.au`

Before asking any questions, make sure you have gone through the materials on Canvas and existing posts on Ed discussion board!

Assignment 2: Reminder

Assignment 2 is now available on Canvas. This assignment contributes up to 10% of your final grade for this subject.

Submission Deadline: Thursday, May 7 at 11:59 pm.

Introduction

- Hypothesis test involving:
  - difference in population proportions
  - independence of two categorical variables
- Simple linear regression

Question 1

Indicate whether each statement below is true or false. Justify your answer.

(a) When testing the difference between two population proportions, a t-test may be used.

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Indicate whether each statement below is true or false. Justify your answer.

(a) When testing the difference between two population proportions, a t-test may be used.

- For testing the difference between two population proportions, i.e. $p_1 - p_2$, we use the Z-statistic which is approximately standard normally distributed.
- FALSE

Question 1

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- Degrees of freedom: $\nu = (r-1) \times (c-1)$
- Since we have 5 rows and 2 columns, $r = 5$ and $c = 2$.

$$\therefore \nu = (5-1) \times (2-1) = 4$$

- TRUE

Question 1

(c) A right-tailed area in the chi-squared distribution equals 0.01. For 4 degrees of freedom, the table value equals 13.2767.

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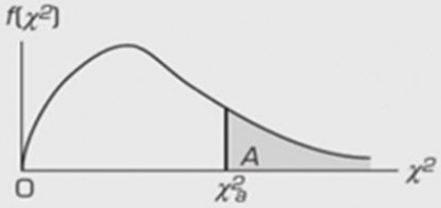
Table 5 Critical Values of the χ^2 Distribution



Degrees of Freedom	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000039	0.000157	0.000982	0.00393	0.0158	2.71	3.84	5.02	6.63	7.88
2	0.0100	0.0201	0.0506	0.103	0.211	4.61	5.99	7.38	9.21	10.6
3	0.072	0.115	0.216	0.352	0.584	6.25	7.81	9.35	11.3	12.8
4	0.207	0.297	0.484	0.711	1.06	7.78	9.49	11.1	13.3	14.9

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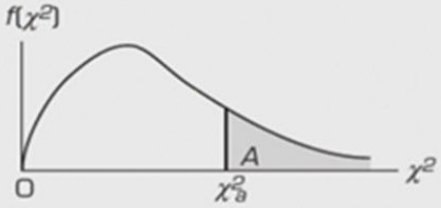

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- $\Pr(\chi^2_4 > 13.2767) \approx \Pr(\chi^2_4 > 13.3) = 0.01$
- TRUE

Question 1

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- Standard deviation of ε (denoted σ_ε) is the same for all values of the regressor (x).
- This is the assumption known as homoscedasticity.
- TRUE

Question 1

(e) A regression analysis between y, sales (in \$1000) and x, advertising (in \$) yielded the least squares line

$$\hat{y} = 60 + 5x.$$

We can interpret the slope by saying that we estimate for each extra \$1 spent on advertising that sales will increase by \$5,000 on average.

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- In a regression model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, $\hat{\beta}_1$ is interpreted as "a unit increase in x is on average associated with a $\hat{\beta}_1$ unit change in y."
- Since the slope coefficient $\hat{\beta}_1 = 5$, x is measured in \$ and y is measured in \$'000, we can interpret for each extra \$1 spent on advertising that sales will increase by \$5,000 on average.
- TRUE

Question 2

A survey to study the usefulness of the online zoom meeting at workplaces used a sample of 1,320 managers, professionals and executives. This group included 528 users of Zoom meetings, 408 people who did not use Zoom meetings and 384 who said they would start to use Zoom meetings relatively soon. Asked if the availability of zoom meetings would boost personal productivity, 50% of the non-users and intenders said no as did 25% of the users.

Can we conclude from these data at the 1% level of significance that users and non-users (including intenders) differ in their opinion of the usefulness of zoom meetings in boosting personal productivity?

Question 2

- Comparison of population proportions
  - What are the two populations?
- Let p_1 be the proportion of users who claim zoom meetings will not boost personal productivity.
- Let p_2 be the proportion of non-users who claim zoom meetings will not boost personal productivity.

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$$H_0 : p_1 - p_2 = 0 \qquad\qquad H_A : p_1 - p_2 \neq 0$$

Contingency table

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	Users	**Non-users**	**Total**
Boost	396	396	792
Doesn't boost	132	396	528
Total	528	792	1,320

Question 2

- Let p_1 be the proportion of users who claim zoom meetings will not boost personal productivity.
- Let p_2 be the proportion of non-users who claim zoom meetings will not boost personal productivity.

Step 1: Is there a difference in their opinion of the usefulness of zoom meetings in boosting personal productivity between users and non-users?

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Step 2: Given independent samples, proportions assumed equal under the null hypothesis, and since $n\hat{p}_1 \geq 5$, $n\hat{q}_1 \geq 5$, $n\hat{p}_2 \geq 5$, $n\hat{q}_2 \geq 5$, then we use:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim N(0,1)$$

Question 2

Step 3: Use $\alpha = 0.01$

Step 4: Reject H_0 if $z < -2.576$ or $z > 2.576$

Standard normal distribution with critical values at $z = -2.576$ and $z = 2.576$.

Excel function: `=NORM.S.INV(0.005)` and `=NORM.S.INV(0.0995)`

Or look up the standard normal cumulative distribution tables.

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<u>Step 5:</u> $\hat{p}_1 = 132/528 = 0.25$, $\hat{p}_1 = 396/792 = 0.5$, $\hat{p} = 528/1,320 = 0.4$ and $\hat{q} = 1 - \hat{p} = 0.6$

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.25 - 0.5}{\sqrt{0.4 \times 0.6 \times \left(\frac{1}{528} + \frac{1}{792}\right)}} = -9.08$$

Question 2

Step 6: Since $-9.08 < -2.576$, we reject H_0 at the 1% level of significance.

There is sufficient evident at the 1% level to support the alternative hypothesis that users and non-users (including intenders) differ in their opinion of the usefulness of zoom meetings in boosting personal productivity.

Question 3

To measure the extent of cigarette smoking around the world random samples of adults in Canada (=1), Australia (=2), New Zealand (=3) and United Kingdom (=4) were drawn. Each was asked whether they smoked (2=Yes, 1= No).

Is there enough evidence to infer that that there are differences in adult cigarette smoking between the four Commonwealth countries? Perform the test at the 5% level of significance.

The survey outcomes resulted in the following numbers of combinations of smoking classification and countries:

$n(1,1) = 165$; $n(1,2) = 167$; $n(1,3) = 164$; $n(1,4) = 156$; $n(2,1) = 35$; $n(2,2) = 33$; $n(2,3) = 36$; $n(2,4) = 44$.

Question 3

$n(1,1) = 165$; $n(1,2) = 167$; $n(1,3) = 164$; $n(1,4) = 156$; $n(2,1) = 35$; $n(2,2) = 33$; $n(2,3) = 36$; $n(2,4) = 44$.

	Canada	**Australia**	**New Zealand**	**UK**	Total
Non-smoker					
Smoker					
Total					

Question 3

Observed frequencies:

	Canada	**Australia**	**New Zealand**	**UK**	Total
Non-smoker	165	167	164	156	652
Smoker	35	33	36	44	148
Total	200	200	200	200	800

Expected frequencies:

	Canada	**Australia**	**New Zealand**	**UK**	Total
Non-smoker					
Smoker					
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Step 2:

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Step 1:

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Step 2:

$$\chi^2 = \sum_{i=1}^{k} \frac{(o_i - e_i)^2}{e_i} \sim \chi^2_\nu$$

where $\nu = (r-1) \times (c-1) = 1 \times 3 = 3$.

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Note: The statistic approximately distributed as χ^2 is ONLY reasonable as long as $e_i \geq 5$ for all i which is satisfied.

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Non-smoker	163	163	163	163	652
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Question 3

Step 3: Use $\alpha = 0.05$

Step 4: Reject H_0 if $\chi^2 > (\chi^2_{0.05,3} = 7.81)$

Step 5: Calculate test statistic

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$$\begin{aligned}\chi^2 &= \frac{(165-163)^2}{163} + \frac{(167-163)^2}{163} + \frac{(164-163)^2}{163} + \frac{(156-163)^2}{163} \\ &+ \frac{(35-37)^2}{37} + \frac{(33-37)^2}{37} + \frac{(36-37)^2}{37} + \frac{(44-37)^2}{37} \\ &= 2.32\end{aligned}$$

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Question 3

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H_0: The two variables are independent

H_A: The two variables are dependent

Step 6:

Since $2.32 < 7.81$, we do not reject H_0 at the 5% level of significance.

There is insufficient evidence to conclude that there are differences in smoking habits between the four Commonwealth countries at the 55 level of significance.

Question 4

Explain what each of the following represents:

(a) $\beta_0 + \beta_1 x + \varepsilon$

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This is the probabilistic relationship between the values of y and x as determined by the true population parameters and a random error term (a population model).

Question 4

(b) $\hat{\beta}_0 + \hat{\beta}_1 x$

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This is the fitted or estimated regression model denoted by $\hat{y}$. The coefficients are the estimated parameters that minimise the sum of squared residuals to create the line of best fit to describe the relationship between the observed values of x and y.

Question 4

(c) $y - \hat{y}$

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(c) $y - \hat{y}$

This is a residual e; the difference between the actual and fitted values of y, and represents variation in y that is not explained by our fitted regression equation.

It is an estimate of the population error ε.

Question 5

A set of 10 observations for the two variables x and y to be analysed by a linear regression model yields the following summations:

$$\sum x = 31 \quad \sum y = 37 \quad \sum xy = 75 \quad \sum x^2 = 103 \quad \sum y^2 = 445$$

Determine the least squares regression line.

Estimating the Regression Line

We use the relationship from Seminar 8:

$$
\begin{aligned}
s_{xy} &= \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \\
&= \frac{1}{n-1} \left[\sum_{i=1}^{n} x_i y_i - n(\bar{x}\bar{y}) \right]
\end{aligned}
$$

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&= \frac{1}{n-1}\left[\sum_{i=1}^{n}x_iy_i - n(\bar{x}\bar{y})\right]
\end{aligned}
$$

Hence:

$$
\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2} \\
&= \frac{\left[\sum_{i=1}^{n}x_iy_i - n(\bar{x}\bar{y})\right]}{\left[\sum_{i=1}^{n}x_i^2 - n(\bar{x}^2)\right]}
\end{aligned}
$$

Your Turn: Calculate the Estimates

Given:

$$n = 10, \qquad \bar{x} = 3.1, \qquad \bar{y} = 3.7, \qquad \sum x_i y_i = 75, \qquad \sum x_i^2 = 103$$

Question: Calculate $\hat{\beta}_1$, $\hat{\beta}_0$, and the estimated regression equation.

Your Turn: Calculate the Estimates

Given:

$$n = 10, \qquad \bar{x} = 3.1, \qquad \bar{y} = 3.7, \qquad \sum x_i y_i = 75, \qquad \sum x_i^2 = 103$$

Question: Calculate $\hat{\beta}_1$, $\hat{\beta}_0$, and the estimated regression equation.

$$\hat{\beta}_1 = \frac{75 - 10(3.1)(3.7)}{103 - 10(3.1)^2} = -5.754$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 3.7 - (-5.754)(3.1) = 21.537$$

$$\hat{y} = 21.537 - 5.754x$$

Thanks for your attention!

Let me know if there is any questions!